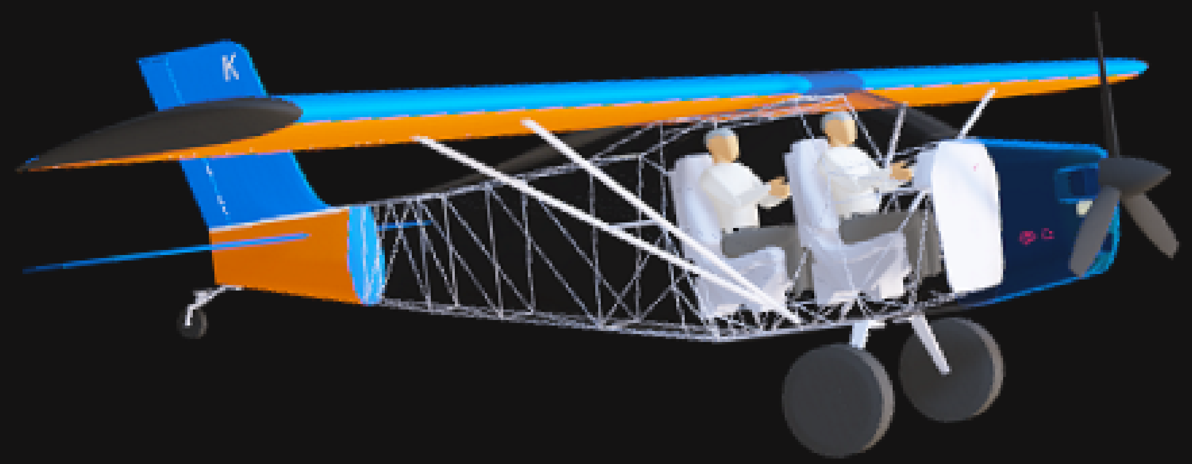
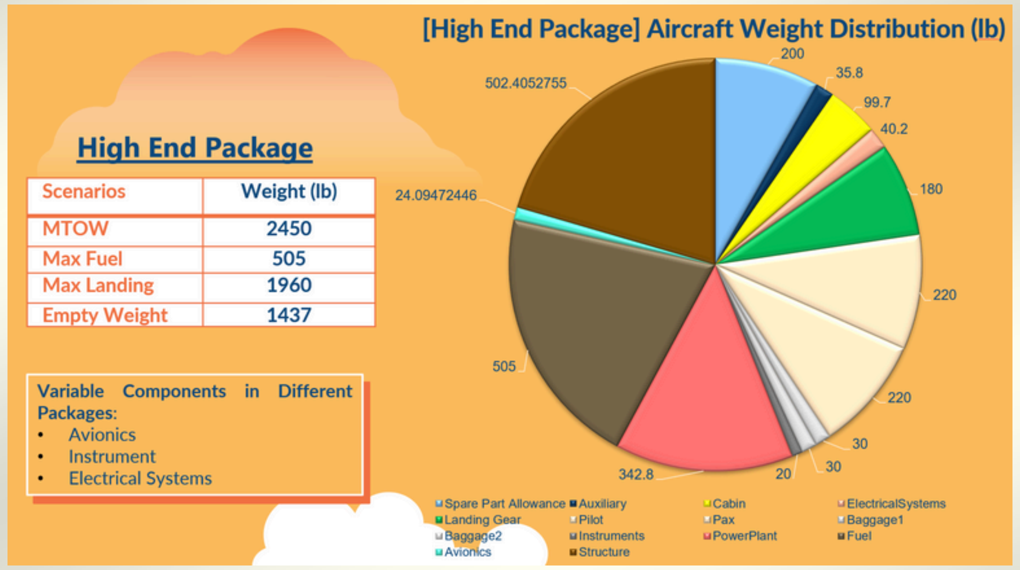
Kit-Plane Design

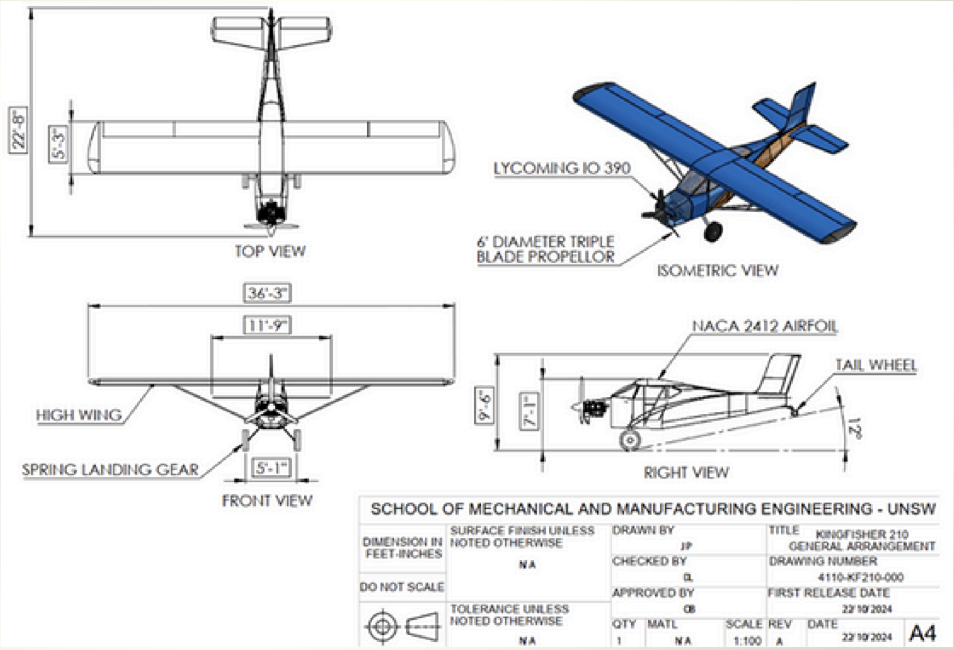
- Responsible for the **design and analysis** on **Weight & Balance**, **Stability**, **Internal layout** of the aircraft, the **Fuselage**, and the **Tail wing AOA**.
- Developed simulations to analyse aircraft flight characteristics, ensuring regulatory compliance with **CASA**, **FAA23** and **EASA** standards.
- Researched and analysed human anthropometric data and ergonomic factors to design the internal layout. Utilized Xfoil, **XFLR5**, **Flow5** and **MATLAB** for 2-3D lift, drag and stability analysis.





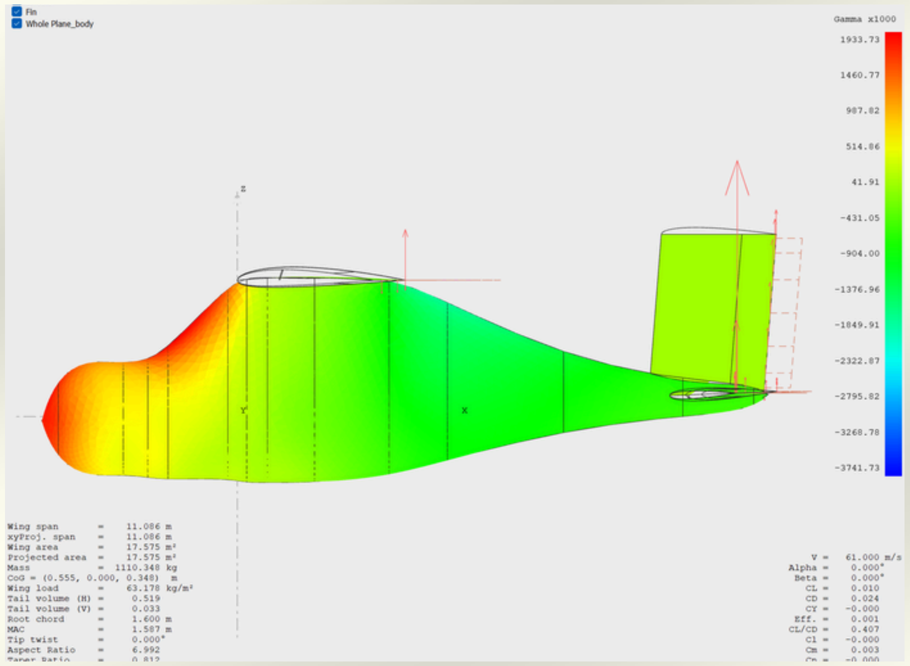
ENGINEERING DRAWING

The final design of the kit-planel



WEIGHT DISTRIBUTION

A detailed weight distribution of aircraft components, estimated from similar aircrafts.



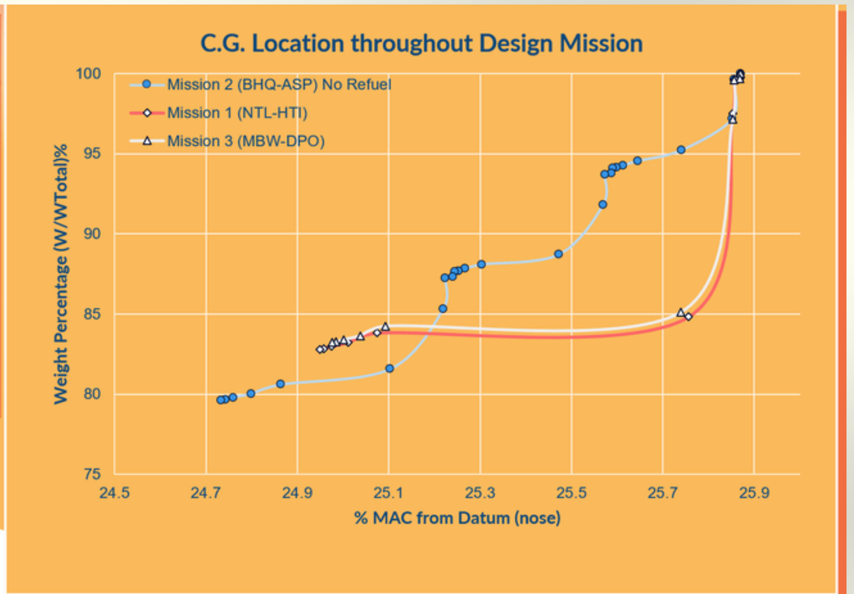
FLOW5 SIMULATION

Simulation of both dynamic and static stability of the aircraft with concatenated lift and drag data of wings from **XFLR5**

CG ENVELOPE

Showcases the shifting of CG throughout each mission. In this case, the CG does not cross aft the neutral point, aircraft remains stable.

CG Positions	Position from nose (in)
Empty Weight CG	104.8"
MTOW CG	107.2"
Forward CG Limit	95" 6 %MAC (When No Equipments are loaded)
Aft CG limit	113.6" 36.1 %MAC
Neutral Point	113.6"
Static Margin	10.2 %MAC



Research Thesis

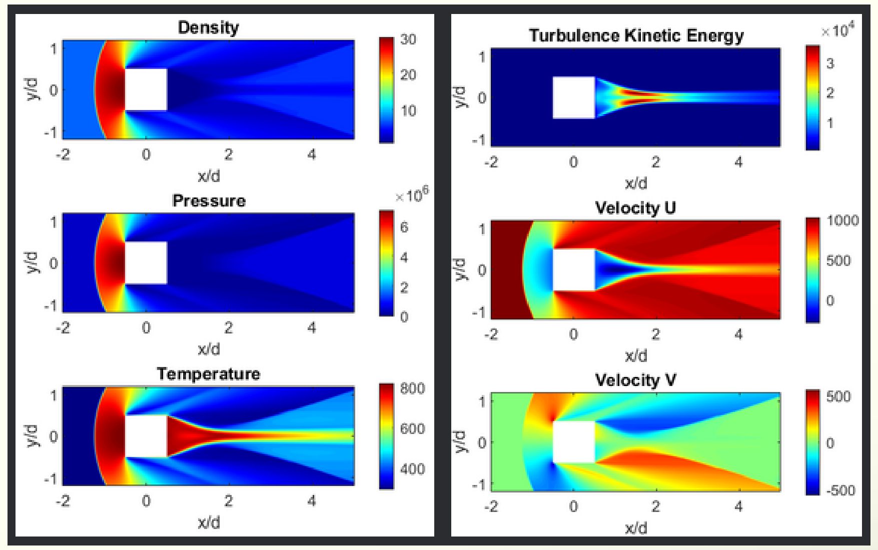
About

This project showcases the simulation of Bluff body near-wake dynamics in supersonic conditions with CFD. Utilizing fluent RANS and DES for flow visualization and MATLAB for plotting and analysis.

Supersonic Bluff Bodies

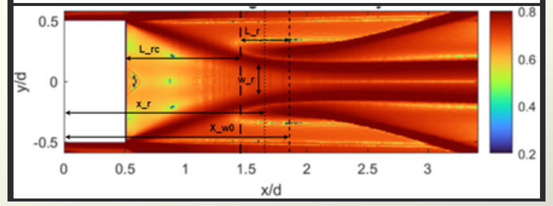
- Aircrafts are subjected to noise generation from many components and environmental aspects. Above subsonic speeds, the complex fluid interactions behind fuel injectors are known to contribute significantly to noise generation.
- The near-wake region of fuel injectors exhibits intricate flow behaviours, including mutual interactions between the distinct flow features and its natural instability. In the process, generating aeroacoustics waves forming a feedback loop that sustains the noise generation as it propagates. Acoustic waves struggle to travel upstream in supersonic conditions. But just so happens the near wake region exhibits a subsonic region due to flow reversal and recirculation.
- This feedback loop excites and reinforces vortex shedding, leading to self-sustained oscillation. With the mutual interactions of other flow feature, this amplification induces a great amount of unsteady loads, which deteriorates the propulsion efficiency and induces structural vibrations and fatigue.

Variables of Interest

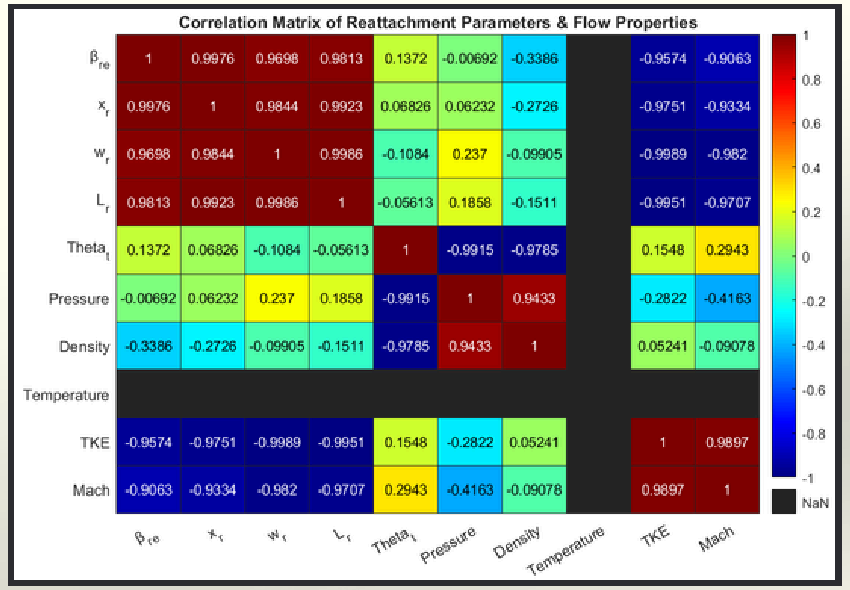


Flow feature length scales:

w_{te}: Trailing edge width
θ_t: Expansion waves turn angle
L_r: Length of reattachment region
L_{rc}: Length of recirculation region
x_r: Streamwise location of the reattachment point
x_{w0}: Streamwise locations of the vortex generation point
w_r: Vertical width of reattachment region
β_{re}: Reattachment Shock Angle

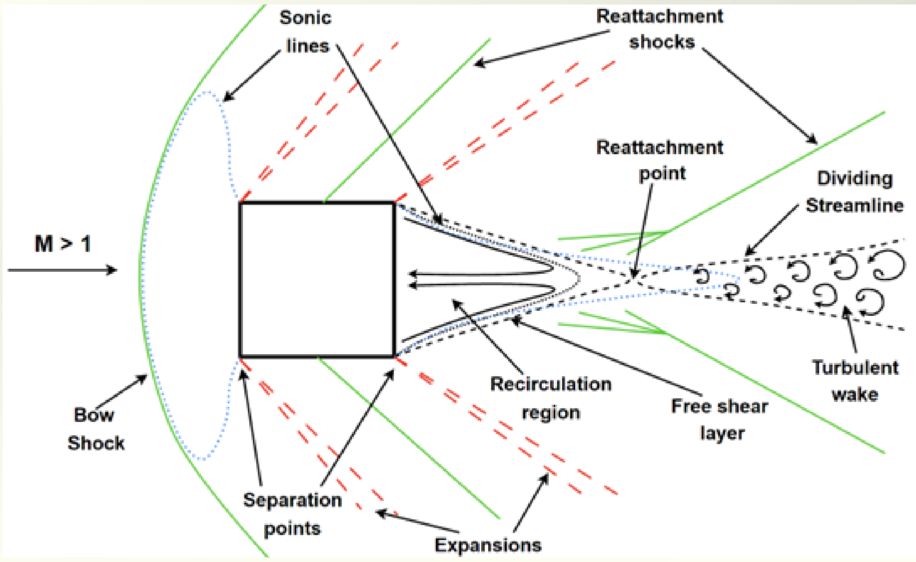


Correlation Matrix



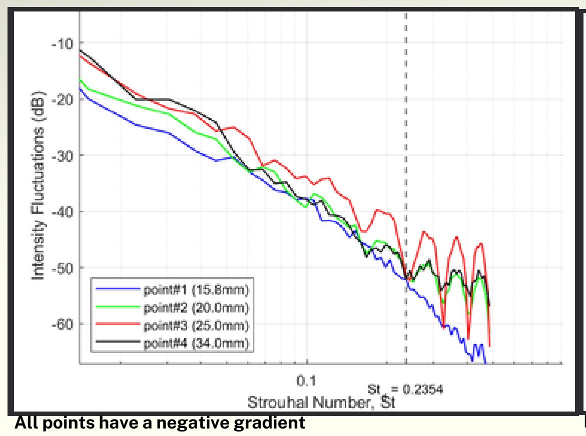
Showcases the distinct flow features in the near wake region. And how each flow properties affect other reattachment parameters in the near-wake region

Square Cylinder in Supersonic Conditions



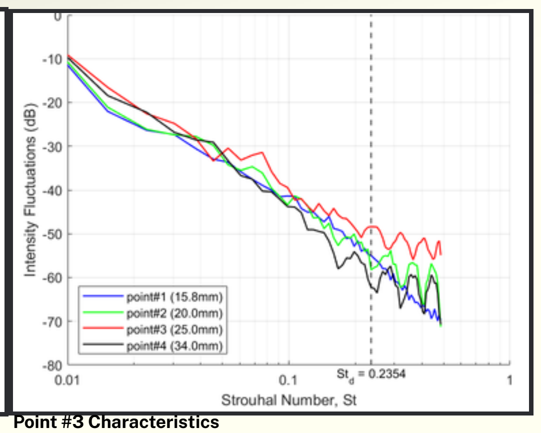
Showcases the distinct flow features in the near wake region

Wake Instability



All points have a negative gradient
Oscillatory behaviour at Point #2 #3 #4

- High oscillatory pressure fluctuations, higher average density fluctuation at point #3
- Point #2 & #4 fluctuations are of the same magnitude
- Point #1 remained at low fluctuations

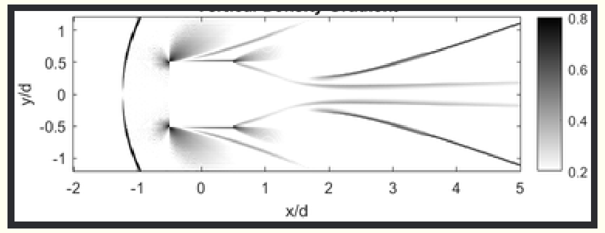


Point #3 Characteristics

- Hump occurred at St = 0.077
- Phase difference of (pi) to #2 and #4
- Lower density flucs due to reattachment shock and the interference of turbulence

These fluctuations are the instabilities within the near-wake region. These pressure fluctuations are also oscillating which confirms the interference of turbulence at the reattachment shock location

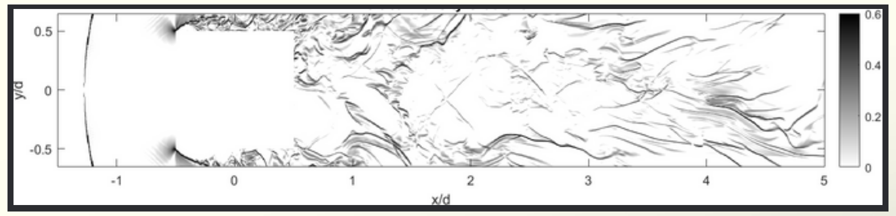
Wake Thickness



k-omega $\delta_w = 1.9671mm$

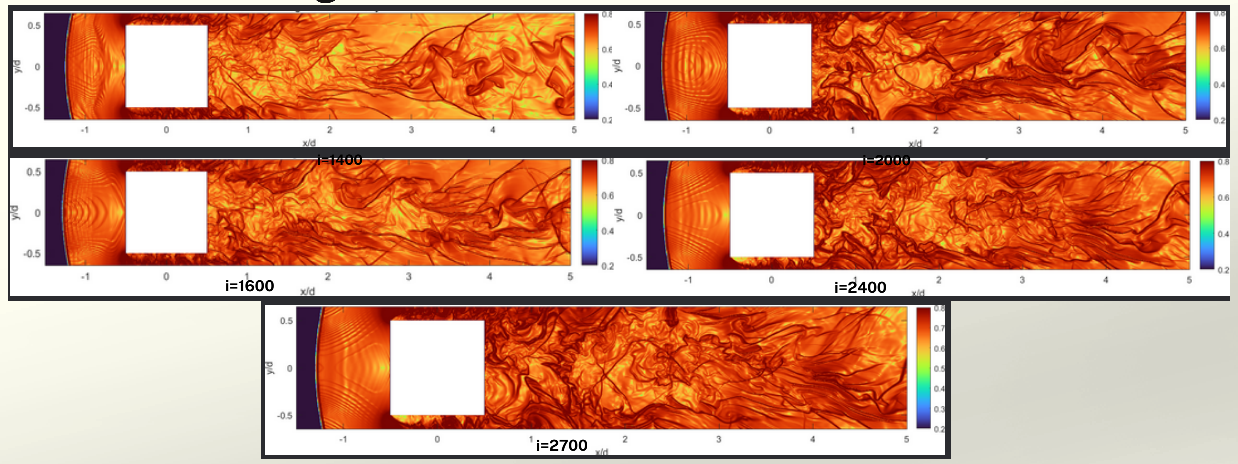
-34.5%

DES $\delta_w = 1.2892mm$



Showcases the reduction in wake thickness between RANS (k-omega) and DES

Schlieren Images (DES)



These are coloured Schlieren images (Density Gradients) at different time-steps. Showcasing the turbulence interference with the reattachment shocks and other distinct features.